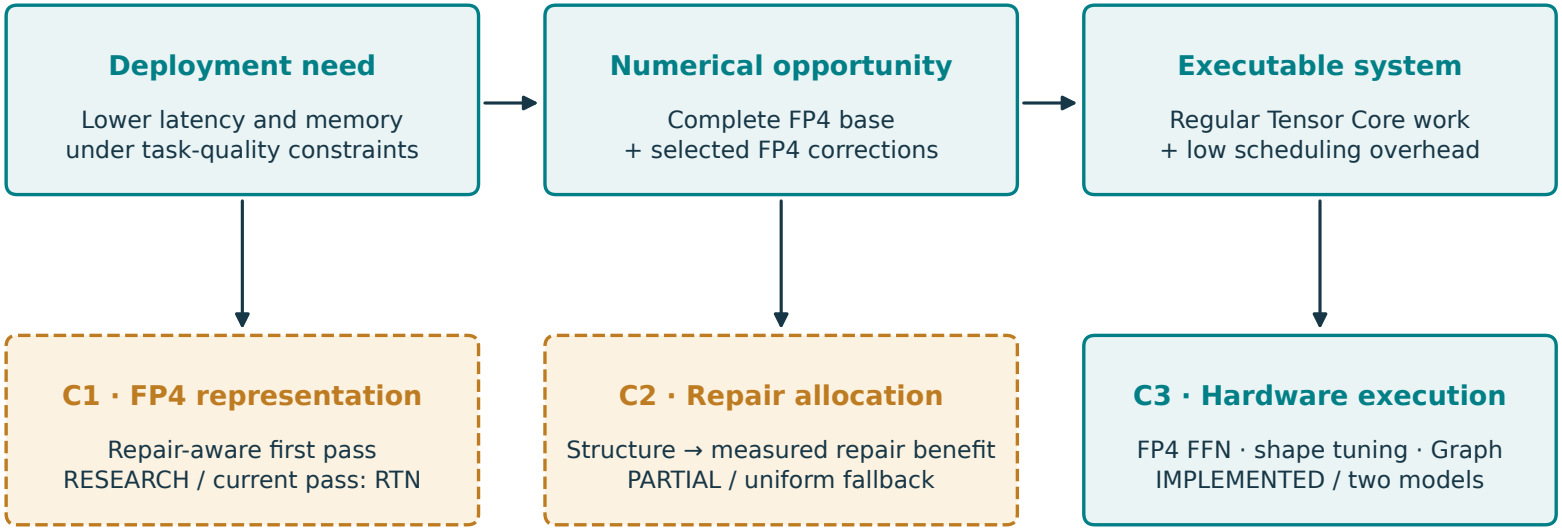
ThunderWorld

Dense FP4 computation + budgeted precision repair

Visual research report · Cosmos3-Edge / Cosmos3-Nano · RTX 5090 · Evidence through 2026-09-11

Executive Summary

Preserve a complete FP4 base; spend extra compute on selected error corrections. The current system implements FFN repair, executable budgets, and equivalent Graph/fusion paths on both models. Repair-aware first-pass quantization and consistently better structural allocation remain research work.



Validation target: complete-trajectory quality × latency × memory; energy remains unmeasured.

Measured: Graph versus each FP4 tier's own eager path: **1.72–2.03× Edge**, **1.16–1.37× Nano**. Execution outputs and budgets match across **120 trajectories**. These are execution gains, not allocator gains or FP4-versus-BF16 speedups. Evidence

1. Overall Architecture

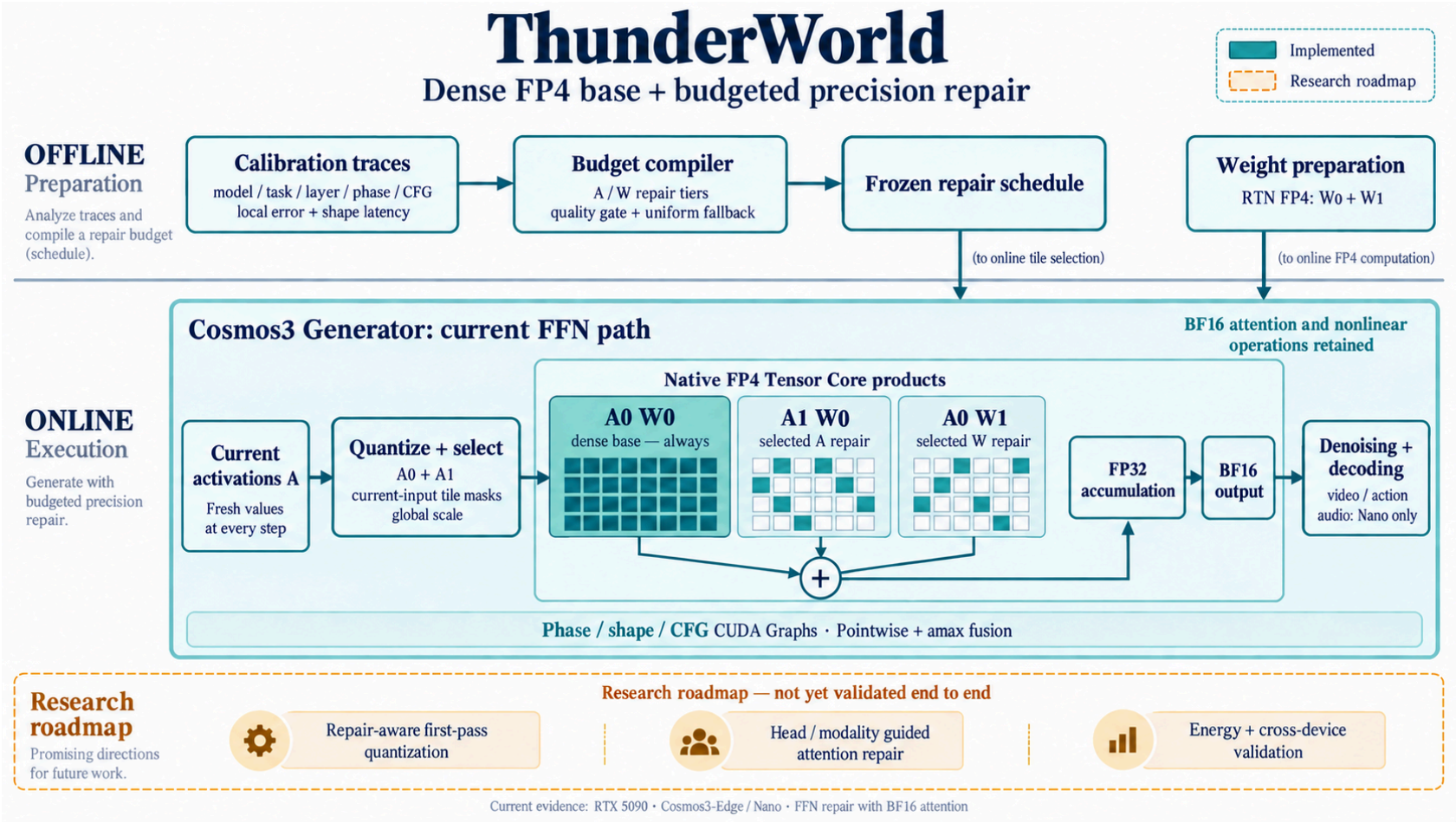
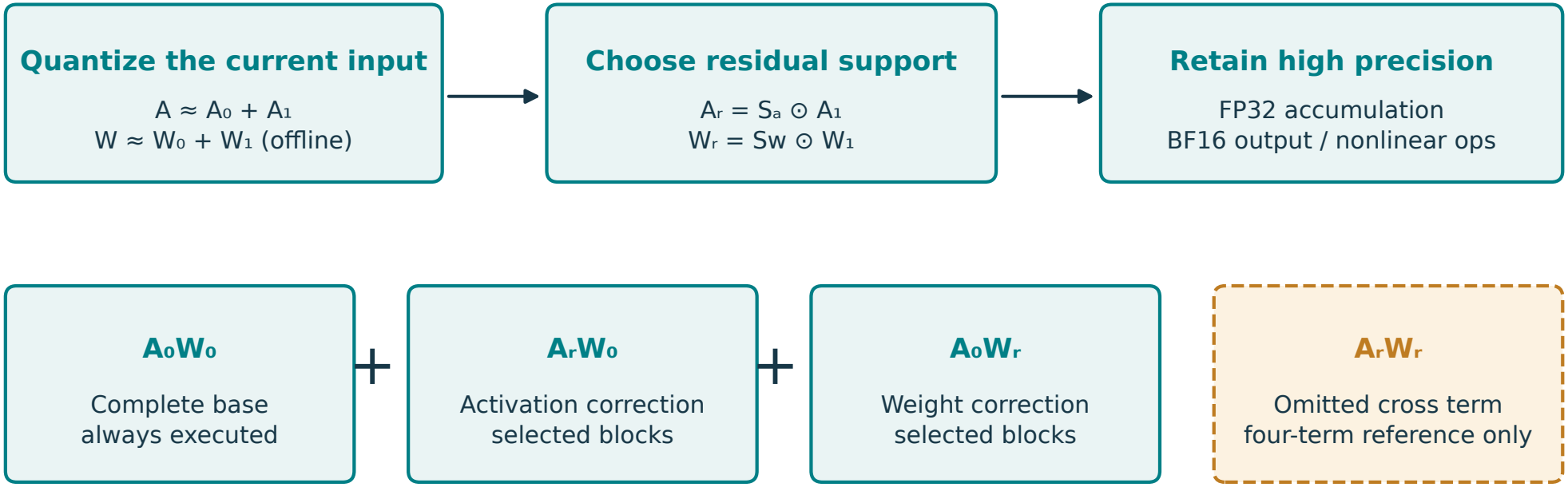


Figure 1. Solid paths are implemented; dashed paths are research directions. The complete base and selected correction products may share a native GEMM. Attention, necessary nonlinear operations, and normalization retain higher precision. Fresh activation values are used every step.

The overview was generated with image_gen; new diagrams and charts in this edition are reproducible vector figures. Image provenance · Detailed report

2. How Precision Repair Works



$$\hat{Y} = A_0W_0 + A_rW_0 + A_0W_r \mid \text{nominal cost} \approx 1 + \rho A + \rho W$$

Logical products may share a native GEMM. FP4 operands are not materialized as complete BF16 matrices.

Figure 2. FP4 codes/scales feed native Tensor Core products. The cross term is omitted in the budgeted first-order path and evaluated separately in the four-term reference.

Numerical gap: the current first pass is absmax RTN. A repair-aware quantizer should optimize the actual selected output, including unrepaired outliers. Two complete FP4 planes use about **9 bit/element** before padding and metadata; that is not fourfold whole-model memory compression. Equations and error identity · Design audit

3. From Calibration to an Executable Budget

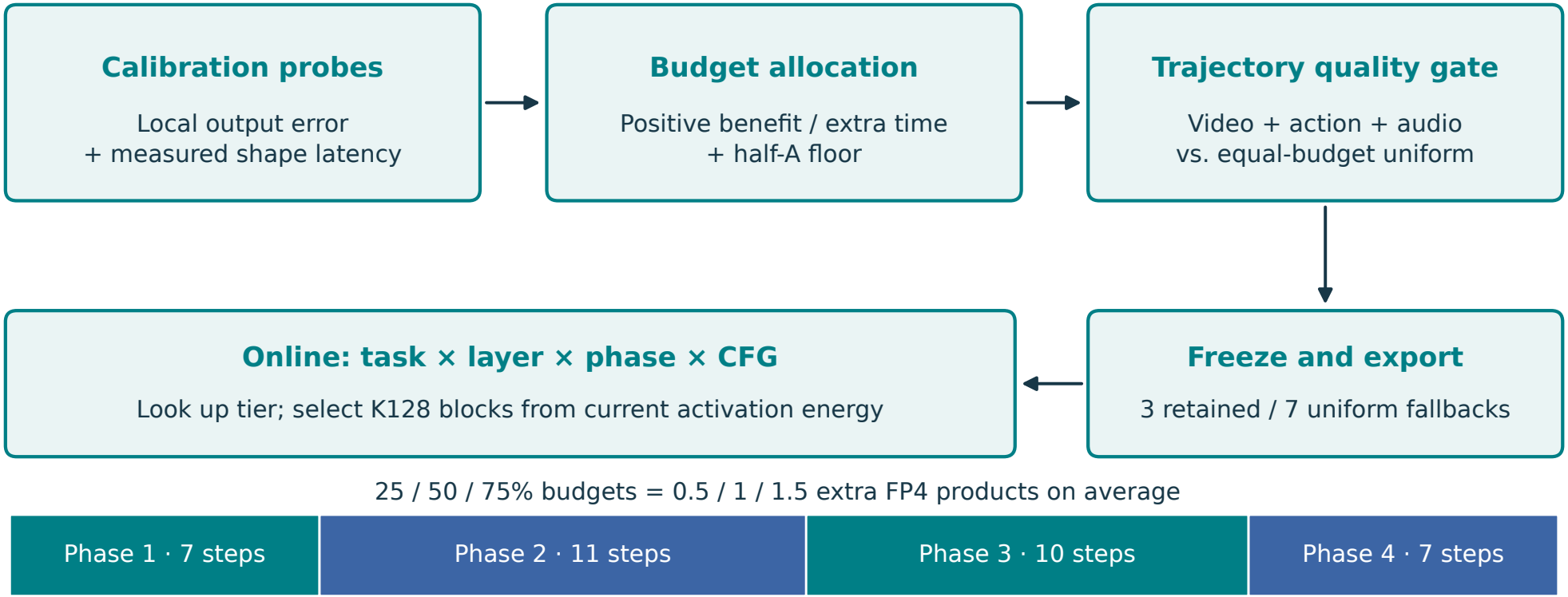


Figure 3. Calibration covers **672 Edge / 2,160 Nano** local units. Tiers vary by task, layer, phase, and CFG branch; half-A support is selected on the GPU from current activations. “Dynamic” does not mean online BF16-teacher evaluation.

Quality gate: only **3 of 10** automatic candidates survive comparison with equal-budget uniform repair; **7 fall back**. With the half-A floor, the 25% tier becomes uniform a50. Local 64-row probes do not fully preserve full-M scale/support context. Allocation evidence

4. Graphs and Fusion Reduce Execution Overhead

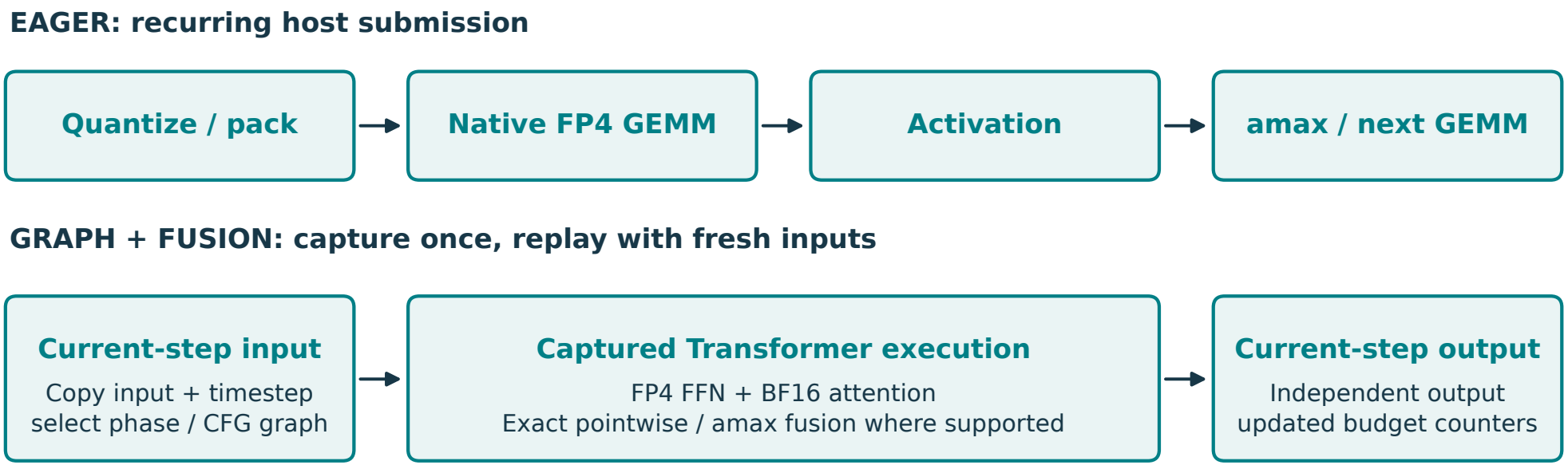


Figure 4. Dependency schematic; box widths do not represent measured duration. Capture is separated by phase/CFG branch and replays with current inputs. Fusion preserves BF16 rounding before amax; Nano SiLU retains its original implementation.

Verified invariant: same outputs and actual repair budgets across eager, Graph, and Graph+fusion for **120 complete trajectories**. This establishes equivalence within a tier, not losslessness against BF16. Runtime and validation

5. Finding A: Deeper DM Layers Attend Less to AR Context

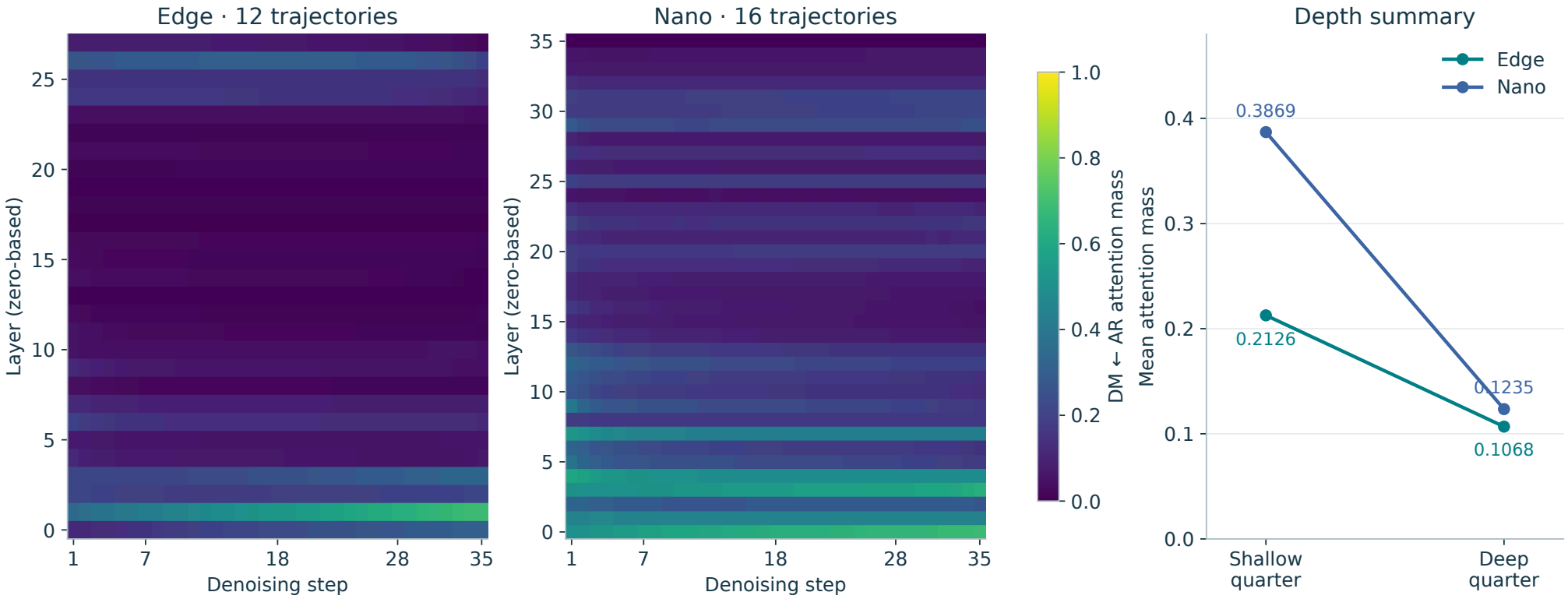


Figure 5. Conditional branch; all 35 steps, layers, and heads; **12 Edge / 16 Nano trajectories**, seeds 6101/6102. DM roles are weighted by query counts; trajectories receive equal weight. Shallow/deep are the first/last layer quarters. The heatmaps share a 0–1 scale.

Implication: depth is a candidate allocation feature, not a reason to delete AR connections. The trend is not monotonic at every layer, and this experiment does not include full AR Reasoner integration. Resampling methodology

6. Finding B: Token Counts Explain Much of the Directionality

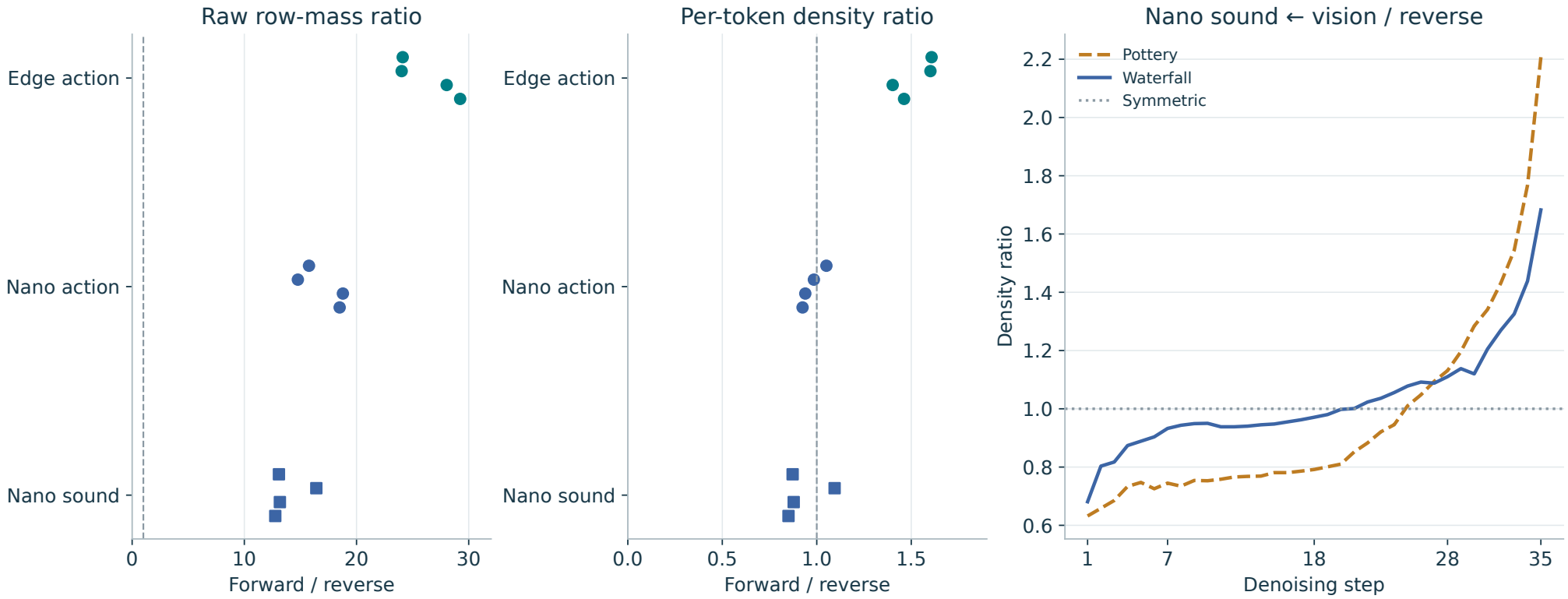


Figure 6. Direction means **modality query ← noisy-vision key**, compared with its reverse. Each dot is one complete trajectory, not a confidence interval. Density ratio is `[m(i,j)/n_j] / [m(j,i)/n_i]`; 1 means symmetry. Nano sound curves average two seeds per scene.

Implication: Edge action retains moderate directionality; Nano's trajectory averages are near symmetric after normalization. Sound directionality changes with step. Raw mass, token counts, and density must all remain visible to the budget model. Definitions and evidence

7. Stable Head Relationships Still Have Changing Outputs

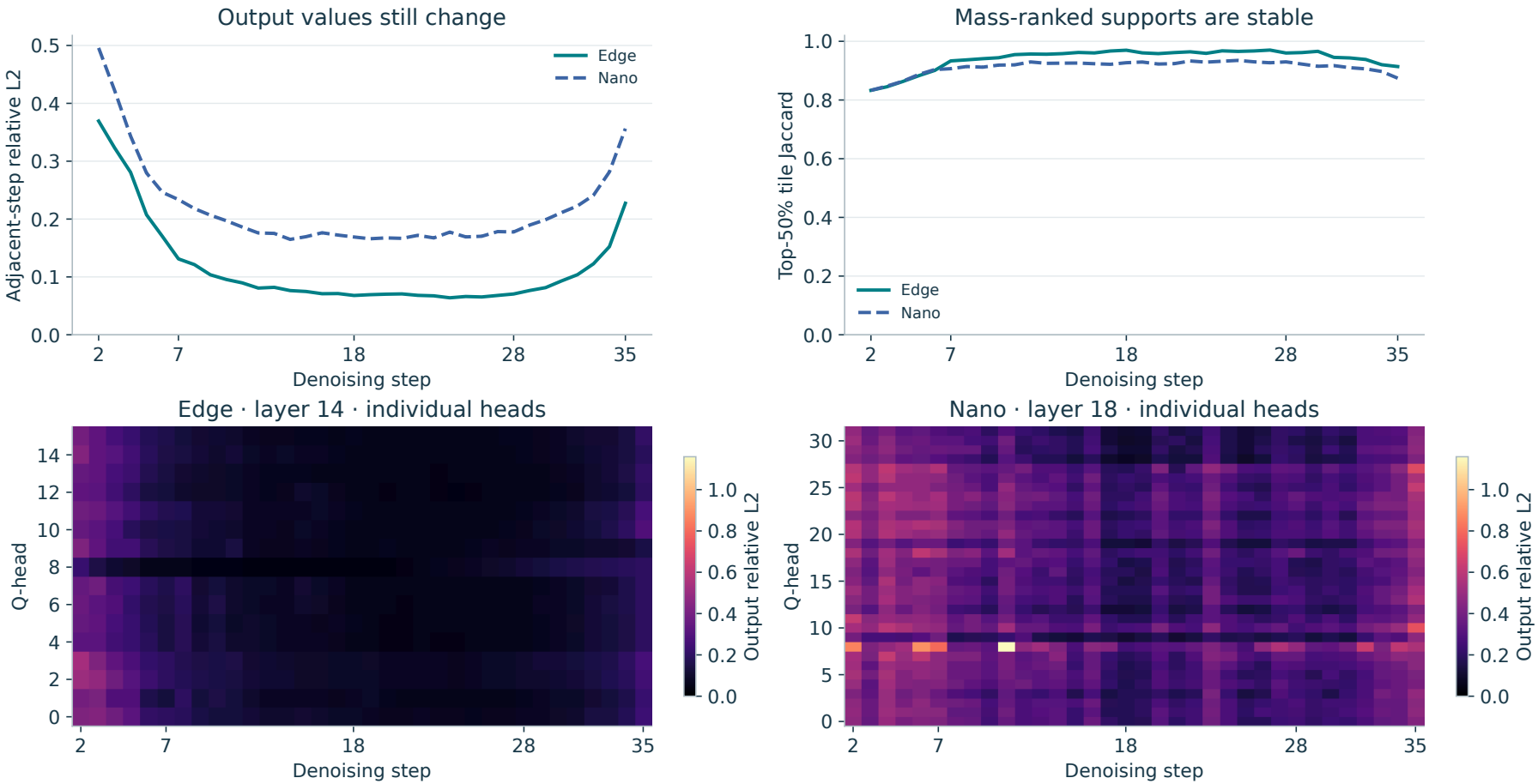


Figure 7. Top: conditional noisy-vision queries averaged over all observation trajectories. Bottom: individual heads at the middle layer, `robot_observe`, mean of two seeds; common heatmap scale. Step 1 is omitted because it has no predecessor. The late-step rise is retained without smoothing.

Implication: dominant roles stay unchanged about **99.16%** of the time, yet mean output relative L2 changes are **0.1163 / 0.2190**. Mass-ranked top-half support is an observation metric, not the runtime repair selector. Head/step analysis

8. Cross-Step Reuse Has Not Delivered a Reliable Win

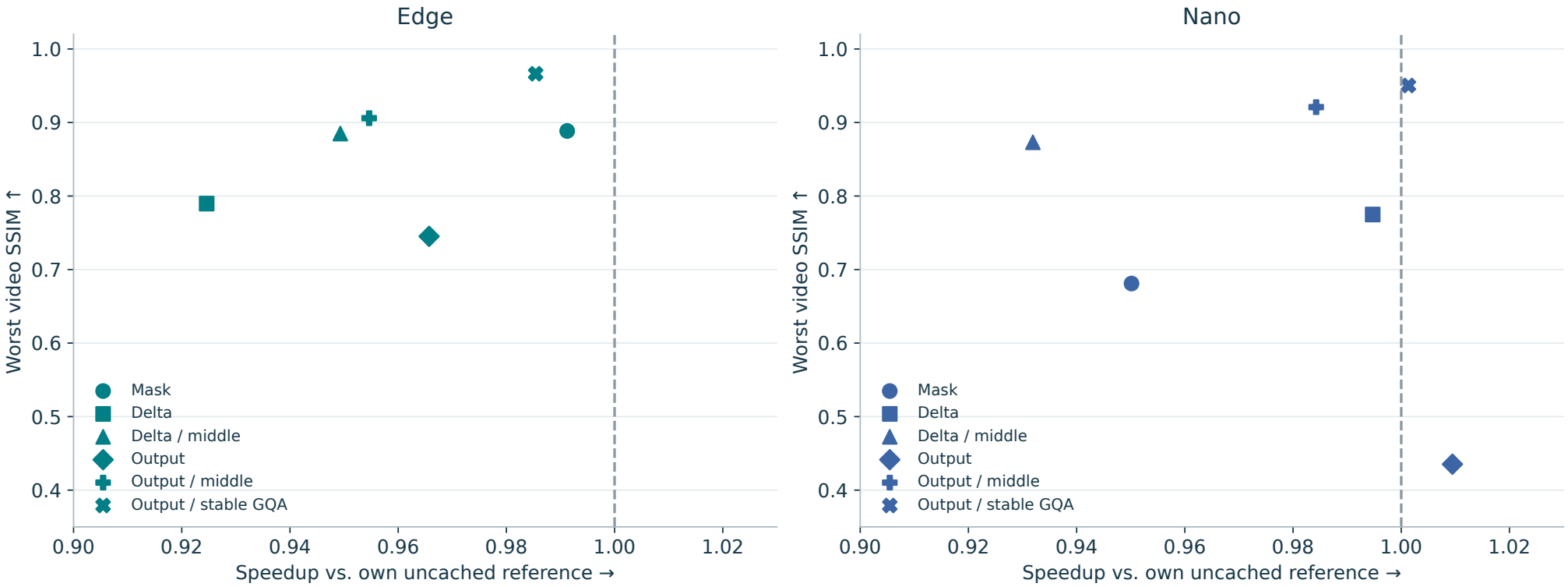


Figure 8. Each point combines worst video SSIM with paired geometric-mean speedup. Compare within each model: their case sets differ. Mask/delta references are uncached attention dual50; output-reuse reference is the FFN ACT2 path with BF16 attention. Seeds 6103/6104; exploratory eager timing.

Decision: keep fresh values in the recommended path. Restricted middle-stage/GQA reuse improves consistency, but **1.001× / 1.010×** single-run timing differences do not establish acceleration. Full-output reuse can reduce Nano AV SSIM to **0.4352** against its uncached reference. Cache ablations

9. Repair Helps, but Its Best Form Depends on the Task

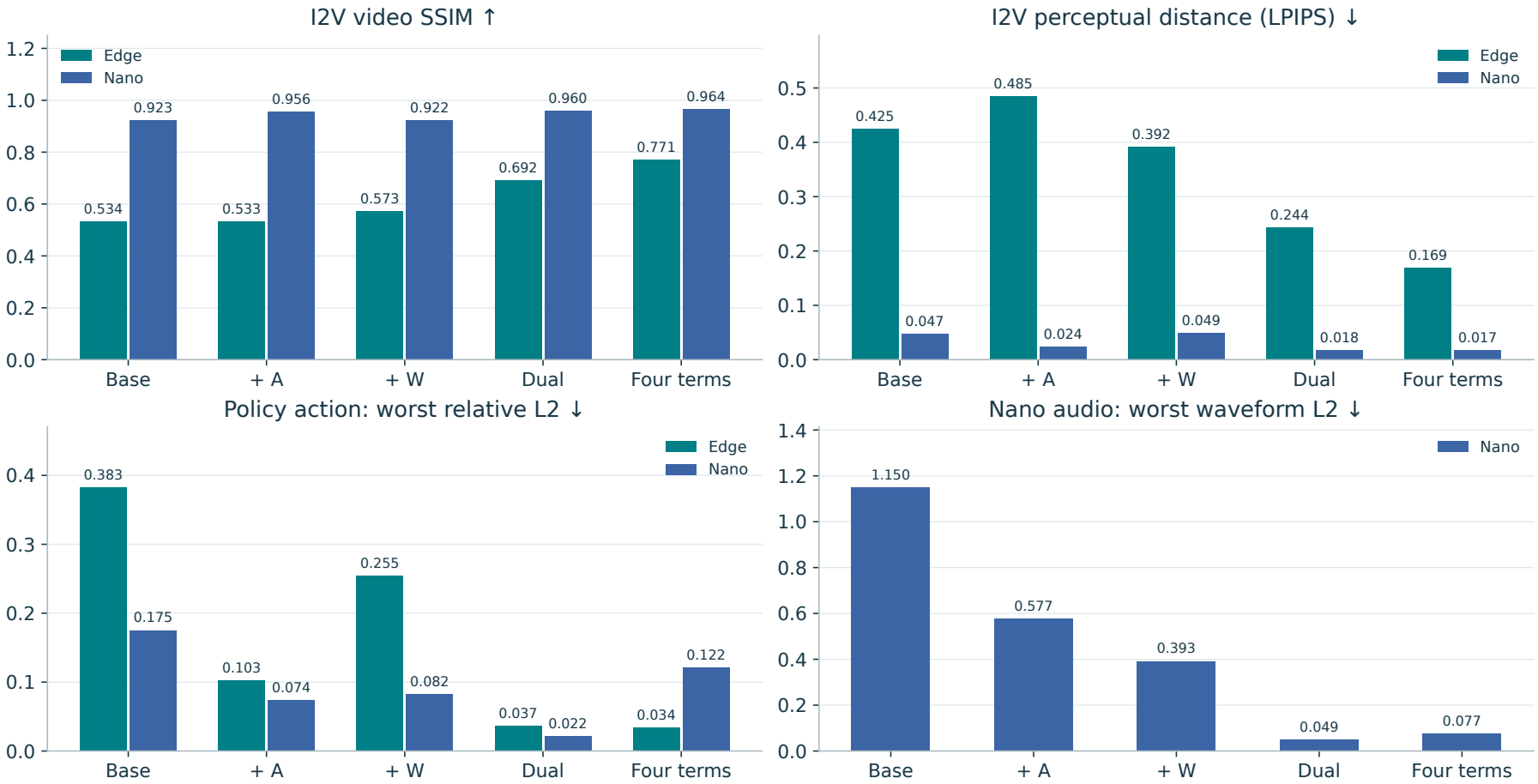


Figure 9. Seeds 6103/6104; all variants retain BF16 attention. I2V SSIM/LPIPS are two-seed means; action and audio show the worst relative L2 across two seeds. All references are same-seed BF16. Edge has no sound path.

Conclusion: dual first-order repair improves several modalities, but adding the cross term can worsen Nano action/audio. A-only is not universally superior to W-only. Edge still has a substantial BF16-consistency gap. These metrics do not measure physical correctness or task success. Complete ablation table

10. End-to-End Speed and Video Consistency

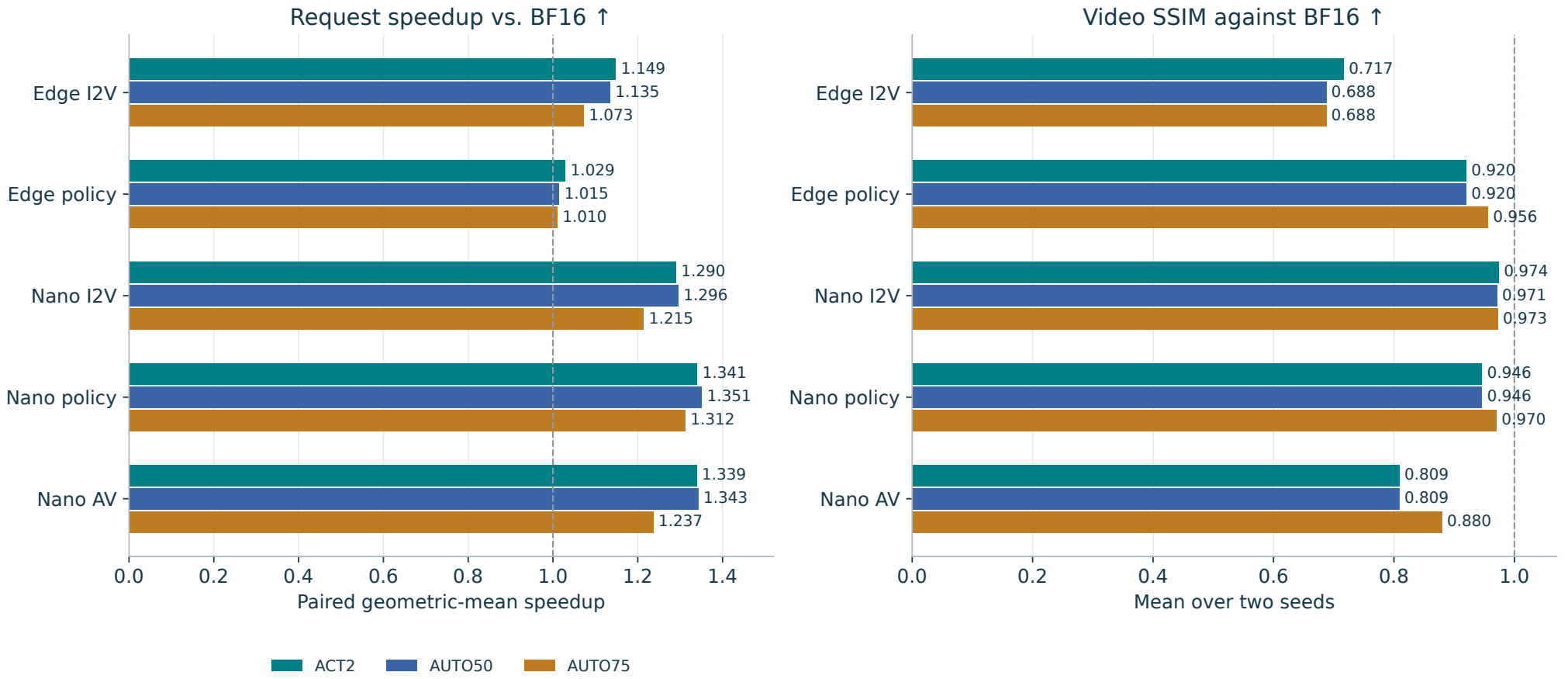


Figure 10. All **15 FP4 operating points**: ACT2, auto50, auto75 across five model/tasks. Seeds 7403/7404; 35 steps; Graph+fusion. BF16 uses the same Graph framework. Speedups are paired geometric means; SSIM is the two-seed mean. Automatic labels include uniform fallbacks.

Timing includes image encoding, denoising, video decoding, and a consistent trajectory sketch; excludes loading, offline quantization, warmup, serialization, AV waveform decoding, and the AR Reasoner. New seeds reuse existing scenes. Action/audio require separate validation. Exact times and all rows

11. Separate Execution Gains from Memory Savings

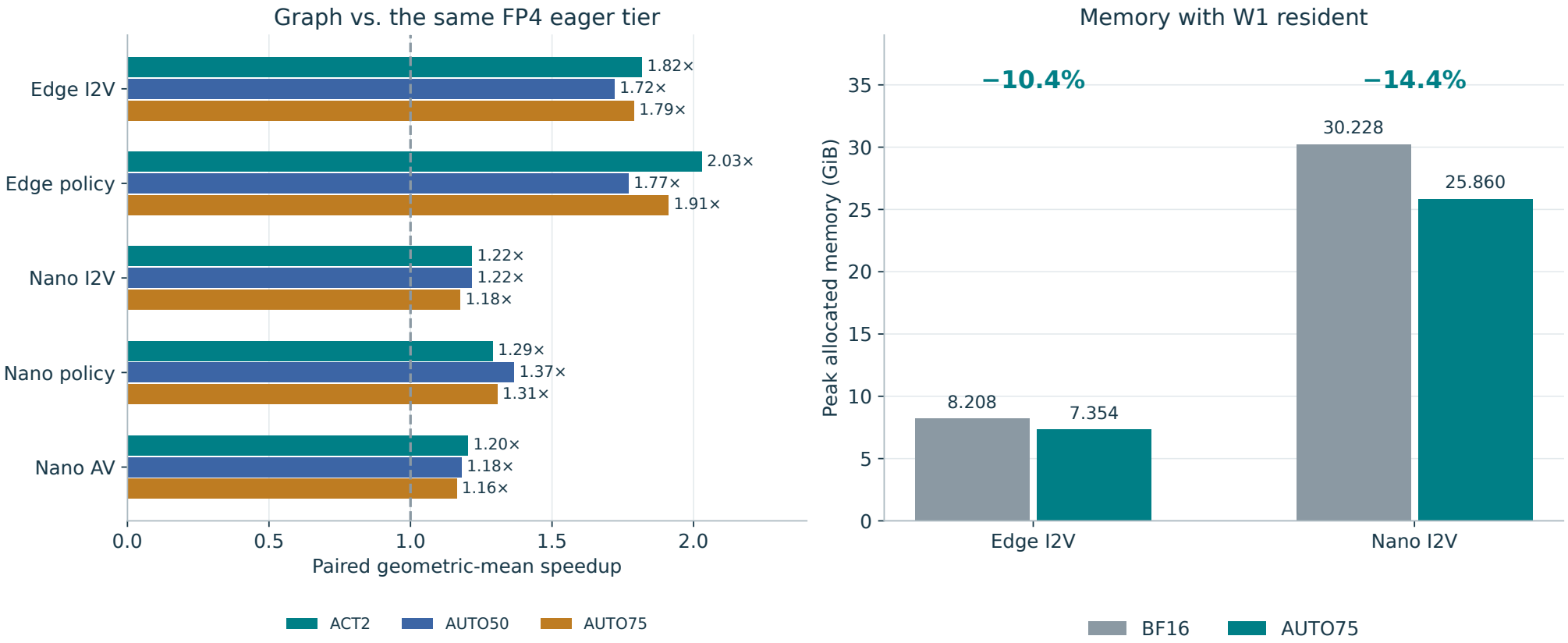


Figure 11. Left: Graph versus the **same FP4 tier's eager path**, not BF16. Right: I2V peak allocated memory, common Graph+fusion framework, W1 resident. Graph adds less than **11 MiB** to peak allocation in the measured configurations.

Pointwise/amax fusion improves local full-FFN time by **3.4–7.2% Edge / 0.2–1.0% Nano**, but its extra request-time benefit is not stable. A BF16 Nano AV negative control changes from **10.992 to 9.233 s** across processes despite no FFN fusion; small gains need repeated timing. Execution evidence

12. Why More FP4 Work or Overlap Is Not Automatically Faster

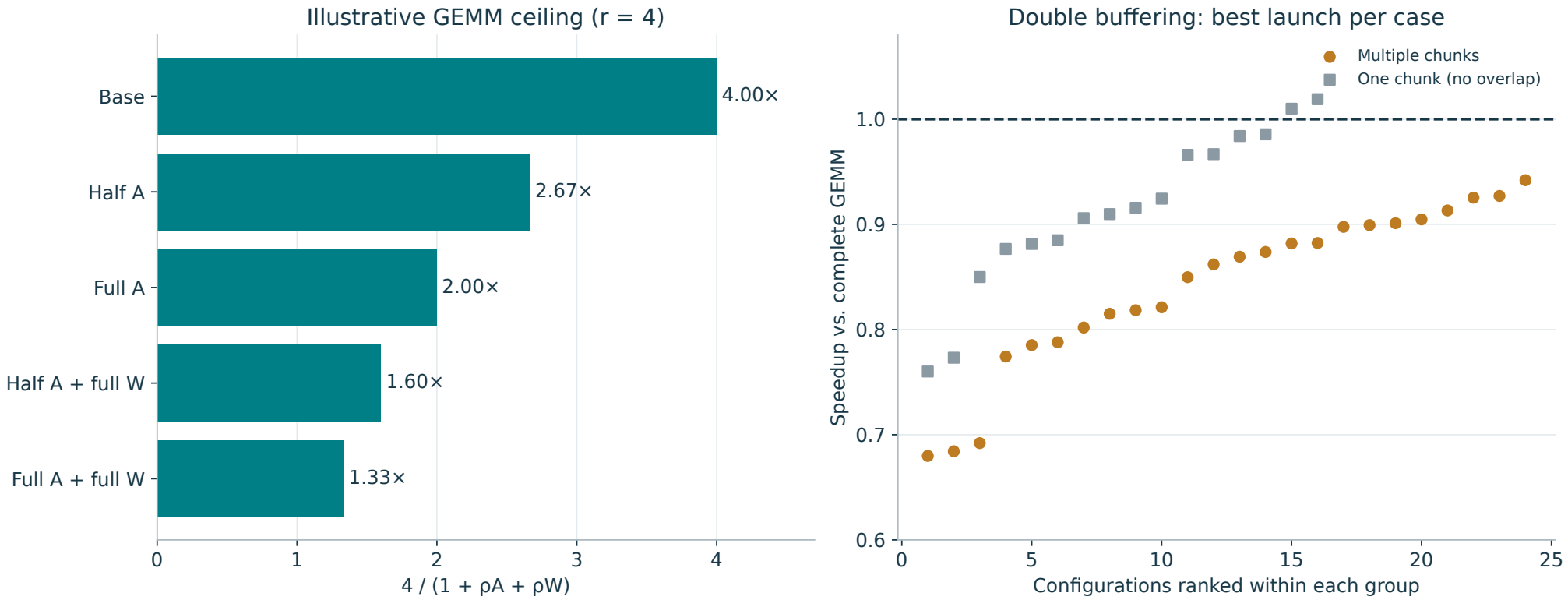


Figure 12. Left is an illustrative compute-only model with FP4/BF16 ratio r=4, not a measured speedup. Right uses **40 retuned cases / 240 launch candidates**, with the best launch selected for each case. One-chunk cases cannot overlap producer and consumer work.

Measured limit: genuine multichunk pipelines remain slower; the best reaches about **0.942×**. Tensor-pipe elapsed activity is **14.2–44.6%**, not peak-FLOP/s utilization. Keep double buffering experimental, preserve K accumulation order, and judge changes by complete-request quality and latency. Pipeline evidence · Profiling

13. What to Complete Next—and What the Demo Should Show

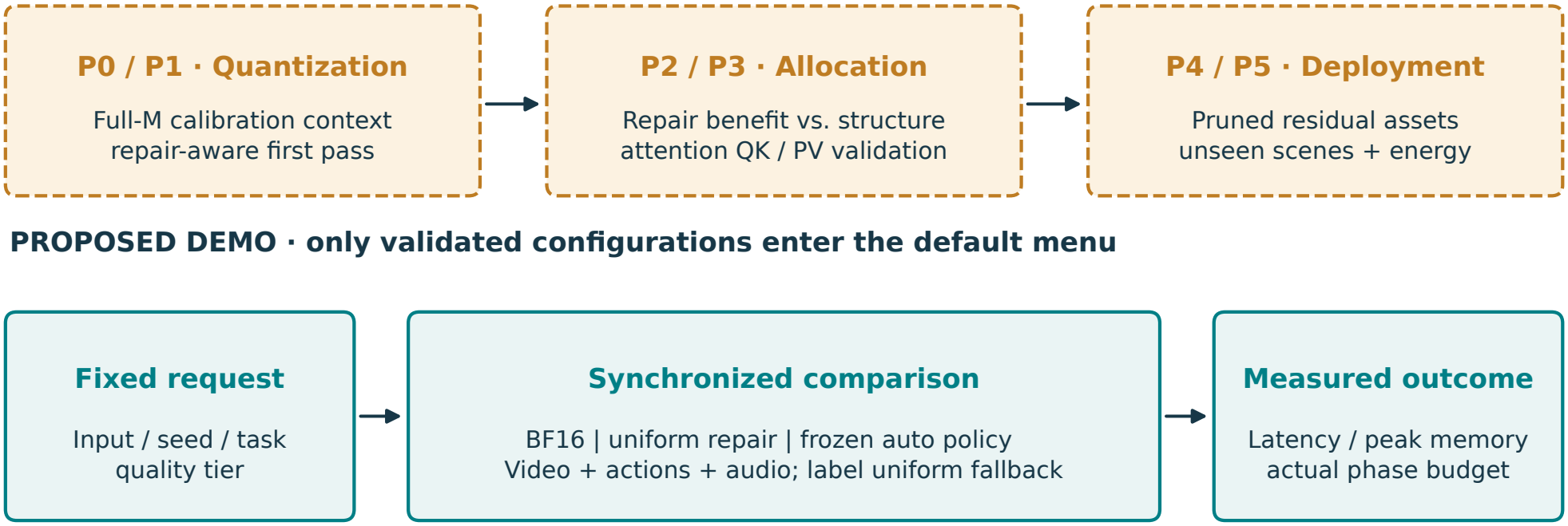


Figure 13. Proposed implementation and demo flow, not an existing UI. The immediate priorities are full-M calibration fidelity, repair-aware first-pass quantization, and structural policies that beat equal-budget uniform/random baselines. Attention requires separate QK/PV calibration and joint-softmax correctness.

Claim boundary: native FP4 FFN execution is validated on RTX 5090; universal structural-allocation gains, near-BF16 quality, reduced energy, and Thor/Spark/Rubin performance remain unproven. Full English report and equations · Detailed PDF · Figure data and source hashes